

**COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE
ASSIGNMENT 4**

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1. Consider a sequence of memory address references given below. In the sequence, each word address is provided in both the decimal and binary formats. Below each address, the relative time at which these references occur is also listed. Memory contents and addresses are shown in the second table.

Data Access Sequence

Time	Address Decimal	Address Binary
✓ 1	28	00 01 11 00
✓ 2	40	00 10 10 00
✓ 3	36	00 10 01 00
4	16	00 01 00 00
5	52	00 11 01 00
6	36	00 10 01 00
7	8	00 00 10 00
8	12	00 00 11 00
9	36	00 10 01 00
10	40	00 10 10 00
11	36	00 10 01 00
12	44	00 10 11 00
13	52	00 11 01 00
14	60	00 11 11 00
15	56	00 11 10 00
16	16	00 01 00 00
17	20	00 01 01 00
18	12	00 00 11 00
19	36	00 10 01 00
20	52	00 11 01 00

Memory

Address Decimal	Address Binary	Data	0	1
8	00 00 10 00	345	1	89
12	00 00 11 00	456	1	200
16	00 01 00 00	4	0	4
20	00 01 01 00	831	3	63
24	00 01 10 00	546	2	34
28	00 01 11 00	65	0	65
32	00 10 00 00	7	0	7
36	00 10 01 00	435	1	179
40	00 10 10 00	9	0	9
44	00 10 11 00	23	0	23
48	00 11 00 00	234	0	234
52	00 11 01 00	97	0	97
56	00 11 10 00	19	0	19
60	00 11 11 00	1000	3	232

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

For fully-associative cache: a word can be stored in any block

Assume that the least-recently used (LRU) scheme is used where appropriate, where the least recently inserted or accessed element is replaced on conflict. Also, when inserting an element into the cache, if there are multiple empty slots for one index, you should insert the new element into the left-most slot (first available slot).

- X a) Use the direct-mapped cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the Tag values. Show the final contents of the cache in the table below, and compute the hit rate and miss rate.

$$\text{Set bits} = 3 \quad C = 8(\text{words}) \quad N = 1$$

$$B = 8 \quad b = \frac{C}{B} = 1 \quad \rightarrow [b = 1 \text{ word}]$$

Index		Cache		
		V	Tag	Data
0	000	0	—	—
1	001	1	001	435
2	010	1	(001) (000)(001)	(9) (345)(9)
3	011	1	(001) (001)(000)	(456) (23)(456)
4	100	1	000	4
5	101	1	(001) (000)(001)	(97) (851)(97)
6	110	1	001	19
7	111	1	(000) (001)	(65) (1000)

Direct-Mapped Cache Summary

$$\text{Hits} = \text{1111} = 6$$

$$\text{Hit Rate: } (6/20) \times 100 = 30\%$$

$$\text{Miss Rate: } (14/20) \times 100 = 70\%$$

$$\text{Misses} = \text{1111 1111} = 14$$

- X b) Use the 2-way set associative cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the Tag values. Show the final contents of the cache in the table below, and compute the hit rate and miss rate.

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

Set	2-way Set Associative Cache							
	V	LRU	Tag	Data	LRU	Tag	Data	V
(00) 0	1	0	(0001)	4	1	—	—	0
(01) 1	1	0	(0010) (0011) (0011)	(435) (831) (97)	1	(0011) (0010)	(97) 435	1
(10) 2	1	0	(0010) (0011)	(9) 19	1	(0000)	345	1
(11) 3	1	0	(0001) (0010) (0000)	(65) (23) 456	1	(0000) (0011)	(456) 1000	1

2-way Set Associative Cache Summary

Hit Rate: $(8/20) \times 100 = 30\%$

Hits = ~~HH~~ I = 6

Miss Rate: $(14/20) \times 100 = 70\%$

Misses = ~~HH~~ ~~HH~~ IIII = 14

- X c) Use the 4-way set associative cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the Tag values. Show the final contents of the cache in the table below, and compute the hit rate and miss rate.

Set	4-way Set Associative Cache											
	V	Tag	Data	V	Tag	Data	V	Tag	Data	V	Tag	Data
0	1	00101	9	1	00010	4	1	00001	345	1	00111	19
1	1	00110	97	1	00111	1000	1	00010	831	1	00100	435

4-way Set Associative Cache Summary

Hit Rate: $\frac{7}{20} \times 100 = 35\%$

Hits = ~~HH~~ II = 7

Miss Rate: $\frac{13}{20} \times 100 = 65\%$

Misses = ~~HH~~ ~~HH~~ IIII = 13

- X d) Use the fully-associative cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the Tag values. Show the final contents of the cache in the table below, and compute the hit rate and miss rate.

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE
ASSIGNMENT 4

Fully Associative Cache

V	Tag	Data	LRU
1	000111 001111	65 1000	011
1	001010 001110	19	010
1	001001 001101	435 831	001
1	000100 001001	4 435	000
1	001101	97	111
1	000010	345	110
1	000011	456	101
1	001011	23	100

Fully Associative Cache Summary

Hit Rate: $\frac{8}{20} \times 100 = 40\%$

Miss Rate: $\frac{12}{20} \times 100 = 60\%$

Hits = 111 111 = 8

Misses = 111 111 11 = 12

2. Fill in the following table for different caches. Assume all caches are direct mapped.

Address Bits	Cache Size	Block Size	Tag Bits	Index Bits	Offset Bits	Bits per Row
16	4KB	4B	4	10	2	37
16	16KB	8B	2	11	3	67
32	8KB	8B	19	10	3	84
32	32KB	16B	17	11	4	146
32	64KB	16B	16	12	4	145
32	512KB	32B	13	14	5	270
64	1048KB	64B	44	14	6	557
64	2048KB	128B	43	14	7	1068

→ 16 + 1648 = 1664

Bits per Row = Tag bits + (Block size × 8) + Value bits

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ASSIGNMENT 4**

3. Assume a direct-mapped cache and data access sequence given below where the memory and its contents are shown in the table on the right. Memory is byte addressed and addresses are given in decimal. Assume five-bit addresses and a one-byte data block.

Data Access Sequence		
Op.	Address	
✓ read	30	
✓ read	31	
✓ read	30	#
✓ read	26	
✓ read	25	
✓ read	28	
✓ read	23	
✓ read	25	#
✓ read	28	
✓ read	20	
✓ read	29	

Cache		
	Tag	Data
0 00	101	5
1 01	111	4
2 10	110	43
3 11	101	13

Memory		
Binary	Address	Data
10100	20	5
10101	21	9
10110	22	47
10111	23	13
11000	24	76
11001	25	6
11010	26	43
11011	27	8
11100	28	12
11101	29	4
11110	30	72
11111	31	101

hits = 11 = 2

hit rate = $\frac{2}{10} \times 100 = 20\%$

misses = 11111 = 8

miss rate = $\frac{8}{10} \times 100 = 80\%$

4. Assume a direct-mapped cache and data access sequence given below where the memory and its contents are shown in the table on the right. Memory is byte addressed and addresses are given in decimal. Assume five-bit addresses and a two-byte data block.

$b = 2$

Offset bits = $\log_2 2$

= 1

Set bits = $\log_2 4$

= 2

Tag bits = $5 - 2 - 1$

= 2

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

Data Access Sequence			
Op.	Address		
read	30	72	Read data
read	31	101	
read	30	72	
read	26	43	
read	25	6	
read	28	12	
read	23	13	
read	25	6	
read	28	12	
read	20	5	
read	29	4	

Cache			
	Tag	Data (0)	Data (1)
0	00	11 76	6
1	01	11 43	8
2	10	11 12 8 12	4 9 4
3	11	11 10 22 47	10 13

Memory		
Binary	Address	Data
10100	20	5
10101	21	9
10110	22	47
10111	23	13
11000	24	76
11001	25	6
11010	26	43
11011	27	8
11100	28	12
11101	29	4
11110	30	72
11111	31	101

hits = 1111 = 4

hit rate = $\frac{4}{11} \times 100 = 36.36\%$

misses = 1111 11 = 7

miss rate = $\frac{7}{11} \times 100 = 63.63\%$

5. Assume a 2-way set associative cache and data access sequence given below where the memory and its contents are shown in the table on the right. Memory is byte addressed and addresses are given in decimal. Assume five-bit addresses and a two-byte data block. You should assume that the replacement policy used here is LRU (least recently used).

Data Access Sequence			
Op.	Address		
lb	30	72	S
lb	31	101	
lb	30	72	
lb	26	43	
lb	25		
lb	28		
lb	23		
lb	25		
lb	28		
lb	20		
lb	29		

Cache			
LRU	Tag	Data (0)	Data (1)
0	101	76 5	8 9
0	111	12	4
1	101	22 47	10 13
1	110	43	8

Memory		
Binary	Address	Data
10100	20	5
10101	21	9
10110	22	47
10111	23	13
11000	24	76
11001	25	6
11010	26	43
11011	27	8
11100	28	12
11101	29	4
11110	30	72
11111	31	101

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

$$\text{hits} = 111 = 5$$

$$\text{hit rate} = \frac{5}{11} \times 100 = 45.45\%$$

$$\text{misses} = 111 - 5 = 6$$

$$\text{miss rate} = \frac{6}{11} \times 100 = 54.54\%$$

6. Consider a sequence of memory access references given below. In the sequence, each address is provided in the decimal format (dec) along with the operation (LDR or STR). Along with each address, the relative time at which these references occur is also listed. Memory contents and addresses are shown in the second table.

Data Access Sequence				Memory		
Time	Op.	Address (dec)	Data (hex)	Address (dec)	Address (bin)	Data (hex)
1	LDR	8		8	00 00 10 00	aa aa aa aa 11 22 33 44
2	LDR	52		12	00 00 11 00	bb bb bb bb 55 66 77 88
3	LDR	12		16	00 01 00 00	cc cc cc cc db db db db
4	LDR	20		20	00 01 01 00	dd dd dd dd
5	STR	16	db db db db	24	00 01 10 00	ee ee ee ee
6	LDR	44		28	00 01 11 00	ff ff ff ff
7	LDR	40		32	00 10 00 00	ca ca ca ca
8	LDR	32		36	00 10 01 00	1a 2b 3c 4d 20 18 20 18
9	STR	60	15 05 15 05	40	00 10 10 00	12 34 56 78 ca ca ca ca
10	STR	36	20 18 20 18	44	00 10 11 00	78 56 4d 12 db db db db
11	LDR	8		48	00 11 00 00	ab cd ef 00
12	LDR	12		52	00 11 01 00	ee 02 04 ee
13	LDR	40		56	00 11 10 00	00 00 00 00
14	LDR	44		60	00 11 11 00	11 11 11 11 15 05 15 05
15	LDR	48		64	01 00 00 00	0f 1e 2d 3c
16	LDR	52				
17	STR	40	ca ca ca ca			
18	STR	44	db db db db			
19	STR	8	11 22 33 44			
20	STR	12	55 66 77 88			

Now consider two different 2-word caches shown on the next page. Assume that each of the two caches are used independently to facilitate memory access for the sequence above. For each cache type, assume that the cache is initially empty.

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

Assume that the least-recently used (LRU) scheme is used where appropriate, where the least recently used or accessed element is replaced on conflict. Also, when inserting an element into the cache, if there are multiple empty slots for one index, you should insert the new element into the first available slot.

For parts (a) and (b) assume the initial memory contents are the ones given above. Update memory in the space above for both parts (try to distinguish between the two if needed.)

- X a) Use the direct-mapped cache shown below with a block size of two words to facilitate memory access the access sequence above. For write hits and misses use the **write back** and **write allocate** policies respectively. You should fill in the binary form of the tag values and hex values for data. When you replace any content in the cache, simply put a cross sign (X) to overwrite a value, and add the new value in the available space. Show the final contents of the memory and cache below, and compute the hit and miss rates.

Hit rate = 10%

Cache Miss rate = 90%

	D	V	Tag	Data 0	Data 1
(00) 0	0	1	0001 0010 0011	db db (caca) abed	db db (caca) ef00
(01) 1	0	1	0011 0001 0010 0011	ee 02 dd dd 2018 ee 02	04 ee dd dd 2018 04 ee
(10) 2	01 0	1	0000 0010 0000 0000	aa aa (1234) 0000 1234 caca 1122	aa aa (5678) 0000 5678 caca 3344
(11) 3	01 0	1	0000 0010 0011 0000 0010 0000	bb bb (2858) (1505) bb bb 2858 dbdb 5566	bb bb (3412) (1505) bb bb 3412 dbdb 7788

- X b) Use the 2-way set associative cache shown below with a of block size two words to facilitate memory access for the access sequence above. For write hits and misses use the **write through** and **write around** policies respectively. You should fill in the binary form of the tag values and hex values for data. When you replace any content in the cache, simply put a cross sign (x) to overwrite a value, and add the new value in the available space. Show the final contents of the memory and cache below, and compute the hit and miss rates.

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE
ASSIGNMENT 4

Hit rate = 15%
Miss rate = 85%

Cache

LRU	Tag	Data 0	Data 1	V
1	0001 0010	aaaa caca 1234 caca	aaaa caca 5678 caca	1
2	0010 0001	1234 aaaa abcd	5678 aaaa ef00	1
3	0010 0001	2002 1111 bbbb 5566	2400 1111 bbbb 7788	1
4	0001 0010	bbbb 7856 ef02	bbbb 3412 0400	1

- c) Which of the above two caches performs better and why? The second one is better because it has more hits than cache no. 1.

7. Show the final data and tags of the direct-mapped cache shown below for the given memory contents and data access sequence. As shown, you have 8-bit addresses and 4 bytes of data in each block. You can assume write-through and write-allocate write strategies on write hits and misses respectively.

Data Access Sequence

Op.	Address	Data
M ✓ lw	00 01 11 00	
M ✓ lw	00 10 10 00	
M ✓ sw	00 10 01 00	32
M ✓ lw	00 01 00 00	
M ✓ sw	00 11 01 00	642
H ✓ lw	00 10 10 00	
H ✓ lw	00 01 11 00	
M ✓ sw	00 10 01 00	124
M ✓ lw	00 10 00 00	
H ✓ lw	00 10 10 00	

Direct-Mapped Cache

V	Tag	Data
	0001 0010	47 43
	0010 0011 0010	32 642 124
	0010	873
	0001	86

Memory

Address	Data
00 00 10 00	435
00 00 11 00	9
00 01 00 00	17
00 01 01 00	13
00 01 10 00	7586
00 01 11 00	86
00 10 00 00	43
00 10 01 00	124
00 10 10 00	673
00 10 11 00	1
00 11 00 00	972
00 11 01 00	103642

hits = 3

hit rate = $\frac{3}{10} \times 100 = 30\%$

misses = 7

miss rate = $\frac{7}{10} \times 100 = 70\%$

- ✓ 8. Fill the following, show final cache

Block size = 2 B

$$N = \frac{8}{2} = 4$$

Assoc =
4-way associative

$$S = 8192 \text{ B}$$

$$b = \frac{31^4}{81} \times 2 = 4 \times 2 = 8 \text{ B}$$

$$B = \frac{8192}{8} = 1024 \text{ B} = 1 \text{ KB}$$

$$S = \frac{B}{1} = 1024$$

$$10. \quad N = 4$$

$$C = 16384 \text{ B}$$

$$b = B \times 4 = 32 \text{ B}$$

$$B = 512$$

$$S = \frac{512}{4} = 128$$

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Address	Memory	
10100	20	7
10101	21	3
10110	22	27
10111	23	32
11000	24	101
11001	25	78
11010	26	59
11011	27	24
11100	28	56
11101	29	87
11110	30	36
11111	31	98

Cache

Address Tag	Data		LRU
	0	1	
11000	101	78	0001
101	7	3	00
110	59	24	00

Processor

1. Read 24 ✓
2. Read 25 ✓
3. Read 26 ✓
4. Read 24 ✓
5. Read 21 ✓
6. Read 26 ✓
7. Read 24 ✓
8. Read 26 ✓
9. Read 27 ✓

$$\text{hits} = 1111 = 6$$

$$\text{hit rate} = \frac{6}{9} \times 100 = 66.67\%$$

$$\text{misses} = 111 = 3$$

$$\text{miss rate} = \frac{3}{9} \times 100 = 33.33\%$$

9. Assume a direct-mapped cache of size 8KB, each block having two 32-bit words. The system is based on 32-bit physical address and uses byte addressing. Find the total number of offsets, index and tag bits.

$$\text{Off Set Bits} = 3$$

$$\text{Index Bits} = 10$$

$$\text{Tag Bits} = 19$$

10. Assume a 4-way set associative cache of size 16KB, each block having four 32-bit words. The system is based on 32-bit physical address and uses byte addressing. Find the total number of offsets, index and tag bits.

$$\text{Off Set Bits} = 5$$

$$\text{Index Bits} = 7$$

$$\text{Tag Bits} = 20$$

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

11. Show final cache where block size = 2, N = 4

Memory			Cache					
Address				Address-Tag	Data			
10100	20	7	L R U			✓ 1	Read 30	✓ 10
10101	21	3				✓ 2	Read 31	✓ 11
10110	22	27	00	11	101 78	✓ 3	Read 30	✓ 12
10111	23	32	00			✓ 4	Read 26	13
11000	24	101				✓ 5	Read 25	14
11001	25	78	1	11	59 24	✓ 6	Read 28	15
11010	26	59	01			7	Read 23	16
11011	27	24		11	56 87	8✓	Read 25	17
11100	28	56	10			9✓	Read 28	18
11101	29	87	01	11	36 98			
11110	30	36	00	10	27 32			
11111	31	98	11					

hits = 111 111 111 = 13

$$\text{hit rate} = \frac{13}{18} \times 100 = 72.22\%$$

misses = 111 = 5

$$\text{miss rate} = \frac{5}{18} \times 100 = 27.78\%$$

12. Show the final data and tags of the two levels of cache, the given memory contents and data access sequence. Consider 5-bit memory addresses and single byte instructions. Assume that the L1 caches are direct mapped and the L2 cache is fully associative, using LRU for replacement. All caches can place a single byte of data in each block. Also assume that addresses greater than 19 contain instructions whereas addresses smaller than 19 contain data.

**COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE
ASSIGNMENT 4**

Access Seq.		L1 Cache		L2 Cache		Memory	
Op.	Adr.	Tag	Data	Tag	Data	Adr.	Data
MM	20	100 111	8 2A 8 1E 29			14	101
MM	15	111 111		10100	5	15	126
MM	30		17			16	3
MM	16	111				17	86
MM	31			01111	126	18	2
MM	15			11110	29	19	91
MM	20			10000	3	20	5
MM	17					21	66
MM	28			12111	17	22	7
MM	15					23	13
MM	30			10001	86	24	4
				11100	12	25	42
						26	0
						27	76
						28	12
						29	3
						30	29
						31	17

Access Seq.		L1D Cache	
Op.	Adr.	Tag	Data
MM	20	100 111	126
MM	15	111 111	86
MM	30	111 111	126
MM	16		
MM	31		
MM	15		
MM	20		
MM	17		
MM	28		
MM	15		
MM	30		

L1(Instruction) miss rate = $\frac{8}{8} \times 100 = 100\%$

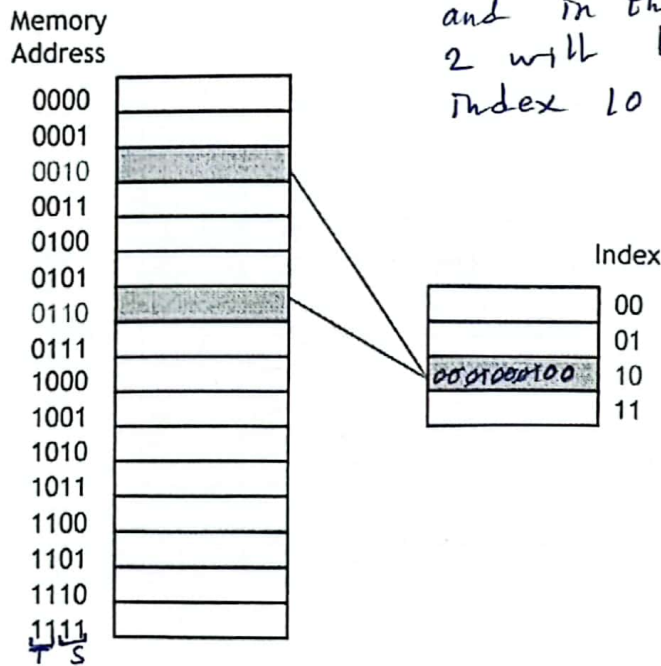
L1(Data) miss rate = $\frac{4}{5} \times 100 = 80\%$

L2 miss rate = $\frac{7}{10} \times 100 = 70\%$

COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE ASSIGNMENT 4

13. If we have a direct mapped cache and memory as below, what happens if a program uses addresses 2, 6, 2, 6, 2?

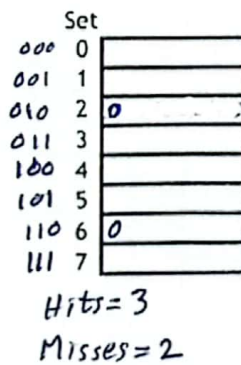
The cache will be missed 5 times and in the end the tag bits of address 2 will be stored in cache at Index 10



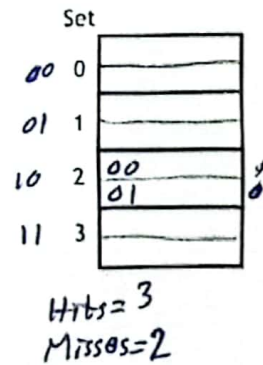
14. Considering the previous question memory and addresses, what happened if we have a fully associative cache?

In case of fully associative cache, the cache will miss two times but not overwrite previous value as if there will be more than one block capacity. And the cache will hit the rest of times.

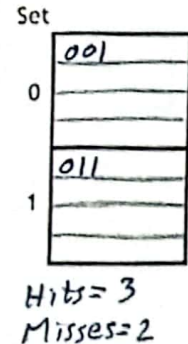
1-way associativity
8 sets, 1 block each



2-way associativity
4 sets, 2 blocks each



4-way associativity
2 sets, 4 blocks each



✓ Q16: Write about the design architecture of the caches used in the following devices and also write a note on why this design architecture is suitable for this device.

1- .386

2- Your Mobile Phone

3- Your Laptop

✗ Q17: Design Cache for direct Mapping? If your Main Memory address size is 7-bits?
Note: Tag is 3-bit, Index/set address is 4 bits, no offset bits

✗ Q18: Design Cache for Set Associative? If your Main Memory address size is 6-bits?
Note: Tag is 3-bits, Index/line is 1 bit, offset is 2-bits, Associativity=4?

✗ Q19: Fill in the Direct-mapped cache for the address given below. Main memory is given on board. Calculate Hit and Miss Rate.

Memory	1	2	3	4	5	6	7	8	9	10	11	4	8
Address	30h	3FH	32H	38H	39H	33H	3EH	31H	10H	1EH	11H	38H	33H
Binary Address	11 000 0	11 111 1	11 001 0	11 100 0	11 100 1	11 001 1	11 111 0	11 000 1	01 000 0	01 111 0	01 000 1	11 100 0	11 001 1
Hit/Miss	M	M	M	M	H	H	H	H	M	M	H	H	H

Tag		0	1	V
11 01	000	1 9	8 11	1
11	001	3	6	1
	010			
	011			
11	100	4	5	1
	101			
	110			
11 01	111	10	2	1

$$\text{Hit Rate} = (7/13) \times 100 = 53.84\%$$

$$\text{Miss Rate} = (6/13) \times 100 = 46.15\%$$

$$\text{Capacity} = 16 \text{ B}$$

$$(\text{block size}) \text{ bytes} = 2 \text{ B}$$

$$B = 8$$

$$N = 1$$

$$S = \frac{8}{1}$$

Q-17:

Add bits = 7

Tag bits = 3

Set bits = 4

off. bits = 0

$$b = 2^0$$

$$\sqrt{b} = 1 \text{ B}$$

$$S = 2^4$$

$$S = 16$$

$$B = 16$$

$$C = B - b$$

$$C = 16 \times 1$$

$$C = 16 B$$

Tag bits	Set/Index	Data
	0000	
	0001	
	0010	
	0011	
	0100	
	0101	
	0110	
	0111	
	1000	
	1001	
	1010	
	1011	
	1100	
	1101	
	1110	
	1111	

Q-18:

Add bits = 6

Tag bits = 3

Set $n = 1$

off. " = 2

$$N = 4$$

$$b = 2^2$$

$$\boxed{b = 48}$$

$$S = 2'$$

$$S \geq 2$$

$$S = \frac{B}{N}$$

$$B = SN = 2(4)$$

$$B = 8$$

$$C = Bb$$

$$= 4 \times 4$$

$$C = 16B$$

		Data	
Tag bits	Index	0	1
	0		
	1		
/	/	/	/
/	/	/	/

Q-16:

-386: Direct Mapped ~~set~~ Architecture

↳ Simpler to implement and sufficient for small sets

Mobile: Multi-level cache (e.g: L1, L2, L3)

L1 → Split data and instructions for faster speed.

L2 → Shared among cores

↳ Mobile need low power, high performance

Laptop: Multilevel cache (L1, L2, Larger L3)

↳ Laptops handle larger workloads and multitasking.

**COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE
ASSIGNMENT 4**

- ✓ 15. Considering the below information:

Technology	Primary Memory	Secondary Memory	Block
Cache Memory	SRAM Cache	DRAM Main Memory	Cache Line
Virtual Memory	DRAM Main Memory	Disk Memory	Page
Access Time	TP (Primary Time)	TS (Secondary Time)	-

Effective Access Time: $T_E = h * T_P + (1 - h) * T_S$, where h (the primary hit rate) is the fraction of memory accesses satisfied by the primary memory; $0.0 \leq h \leq 1.0$.

This formula does extend to multi-level caches. For example, a two-level cache has $T_E = h_1 * T_1 + (1 - h_1) * h_2 * T_2 + (1 - h_1) * (1 - h_2) * T_S$.

- Suppose a single cache fronting a main memory, which has 80 nanosecond access time. Suppose the cache memory has access time 10 nanoseconds.

(a) If the hit rate is 90%, then T_E $= ?$
 $= (0.9 \times 10) + (0.1 \times 80)$
 $= 9 + 8 = 17 \text{ ns}$

(b) If the hit rate is 99%, then T_E $= ?$
 $= (0.99 \times 10) + (0.01 \times 80)$
 $= 9.9 + 0.8 = 10.7 \text{ ns}$

(c) Suppose a L1 cache with $T_1 = 4$ nanoseconds and $h_1 = ?$
 $h_1 = \frac{7}{13} = 0.53$

(d) Suppose a L2 cache with $T_2 = 10$ nanoseconds and $h_2 = ?$

(e) Suppose a main memory with $T_S = 80.0$ then $T_E = ?$
 $= 7.9 \text{ ns}$